# Automatic Supervision of Land Cover Labels to Increment the Reliability of Training Datasets for Post-Disaster Recovery Monitoring

Martin Garcia-Fry[1*], Osamu Murao[2], Bruno Adriano[2], Shunichi Koshimura[2]

## Affiliations

1. Department of Architecture and Building Science, Graduate School of Engineering, Tohoku University, Sendai, Japan.
2. International Research Institute of Disaster Science, Tohoku University, Sendai, Japan.

*Corresponding author: m_garcia15@hotmail.com

## Supplementary Material

This file provides supplementary information complementing results of this study.

Figure S1. Cropland NDVI time-series from March (growth) through October (senescence). Smoothed, mean, and raw amplitudes are compared. Raw metrics outperformed smoothed and mean metrics given their ability to characterize signal variability while avoiding a model to overfit. Further, low levels of sensor-derived noise are seen at the ~80, ~163, ~462, and ~736 markers.

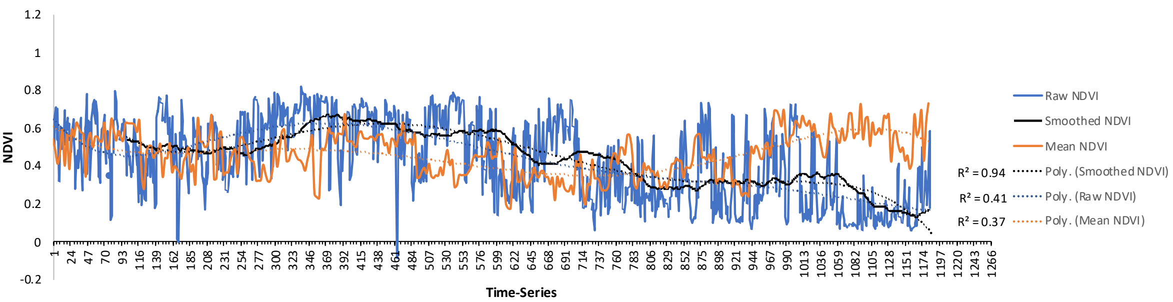


Table S1. Multitemporal metrics for the Indo-Malayan tropical ecozone. The enumerated ranges represent smoothed, mean, and multitemporal raw metrics. Water and wetland samples were constrained to the number of available units and may not cast a significant representation of the ecozone.

| **01_Dense Short Vegetation** | **Samples** | **Min** | **Max** | **Standard Error** | **Range** | |
|---|---|---|---|---|---|---|
| *Smoothed (NIR-Red / NIR+Red)* | 1,192 | 0.254 | 0.742 | 0.004 | *0.258* | *0.738* |
| *Mean* | 323 | 0.255 | 0.755 | 0.006 | 0.261 | 0.749 |
| *Temporal* | 1,292 | 0.085 | 0.858 | 0.005 | *0.090* | *0.853* |
| *Smoothed (Wetness Index)* | 1292 | -3.866 | 8.466 | 0.056 | -3.81 | 8.41 |
| *Mean* | 323 | -3.866 | 8.466 | 0.112 | -3.753 | 8.353 |
| *Temporal* | 1,292 | -3.866 | 8.466 | 0.056 | -3.810 | 8.410 |
| *Smoothed (Solar Radiation)* | 1292 | 4633.647 | 7884.382 | 13.617 | 4647.26 | 7870.77 |
| *Mean* | 323 | 4633.65 | 7884.38 | 27.27 | 4660.91 | 7857.12 |
| *Temporal* | 1,292 | 4633.65 | 7884.38 | 13.62 | 4647.26 | 7870.77 |
| *Smoothed (NIR-SWIR2/NIR+SWIR2)* | 1192 | 0.226 | 0.715 | 0.004 | 0.230 | 0.711 |
| *Mean* | 323 | 0.163 | 0.685 | 0.005 | 0.168 | 0.680 |
| *Temporal* | 1,292 | -0.037 | 0.761 | 0.005 | -0.033 | 0.756 |
| *Smoothed (Tasseled Cap Greenness)* | 1292 | 1.49E-06 | 2.01E-01 | 1.45E-03 | 0.001 | 0.200 |
| *Mean* | 323 | 0.004 | 0.225 | 0.002 | 0.006 | 0.223 |
| *Temporal* | 1,292 | -0.066 | 0.321 | 0.002 | -0.064 | 0.319 |
| **02_Open Tree Cover** | **Samples** | **Min** | **Max** | **Standard Error** | **Range** | |
| *Smoothed (NIR-Red / NIR+Red)* | 2,144 | 0.437 | 0.749 | 0.002 | *0.439* | *0.747* |
| *Mean* | 560 | 0.305 | 0.795 | 0.004 | *0.309* | *0.791* |
| *Temporal* | 2,244 | 0.140 | 0.858 | 0.003 | *0.143* | *0.855* |
| *Smoothed NDWI_I (NIR-SWIR1 / NIR+SWIR1)* | 2,144 | 0.115 | 0.377 | 0.002 | 0.117 | 0.376 |
| *Mean* | 560 | 0.007 | 0.478 | 0.003 | 0.011 | 0.475 |
| *Temporal* | 2,244 | -0.119 | 0.505 | 0.002 | -0.116 | 0.502 |
| *Smoothed (Tasseled Cap Greenness)* | 2,144 | 0.056 | 0.171 | 0.001 | 0.057 | 0.171 |
| *Mean* | 560 | 0.015 | 0.206 | 0.001 | 0.016 | 0.204 |

| | | | | | | |
|---|---|---|---|---|---|---|
| *Temporal* | 2,244 | -0.053 | 0.278 | 0.001 | -0.052 | 0.277 |
| *Smoothed (Solar Radiation)* | 2244 | 3804.301 | 7598.074 | 14.280 | 3818.58 | 7583.79 |
| *Mean* | 560 | 3804.30 | 7598.07 | 25.61 | 3829.91 | 7572.46 |
| *Temporal* | 2,244 | 3804.30 | 7598.07 | 14.28 | 3818.58 | 7583.79 |
| *Smoothed GNDVI (NIR-Green / NIR+Green)* | 2144 | 0.445 | 0.656 | 0.001 | 0.446 | 0.655 |
| *Mean* | 560 | 0.336 | 0.701 | 0.003 | 0.339 | 0.698 |
| *Temporal* | 2,244 | 0.218 | 0.745 | 0.002 | 0.220 | 0.743 |
| *Smoothed (Ground Height)* | 2244 | 20.85 | 979.44 | 4.28 | 25.13 | 975.16 |
| *Mean* | 560 | 20.848 | 979.445 | 8.587 | 29.43 | 970.86 |
| *Static* | 2,244 | 20.848 | 979.445 | 4.283 | 25.13 | 975.16 |
| *Smoothed (Slope)* | 2244 | 0.05 | 33.03 | 0.17 | 0.22 | 32.86 |
| *Mean* | 560 | 0.049 | 33.033 | 0.337 | 0.386 | 32.695 |
| *Static* | 2,244 | 0.049 | 33.033 | 0.168 | 0.217 | 32.864 |
| *Smoothed (NIR-SWIR2 / NIR+SWIR2)* | 2144 | 0.385 | 0.685 | 0.002 | 0.386 | 0.683 |
| *Mean* | 560 | 0.195 | 0.743 | 0.004 | 0.199 | 0.739 |
| *Temporal* | 2,244 | 0.083 | 0.776 | 0.003 | 0.086 | 0.773 |
| **03_Dense Tree Cover** | **Samples** | **Min** | **Max** | **Standard Error** | **Range** | |
| *Smoothed (NIR-Red / NIR+Red)* | 1060 | 0.558 | 0.731 | 0.002 | *0.560* | *0.730* |
| *Mean* | 290 | 0.314 | 0.809 | 0.005 | *0.320* | *0.804* |
| *Temporal* | 1,160 | 0.140 | 0.852 | 0.003 | *0.143* | *0.848* |
| *Smoothed (NIR-SWIR2 / NIR+SWIR2)* | 1060 | 0.506 | 0.697 | 0.002 | 0.508 | 0.695 |
| *Mean* | 290 | 0.250 | 0.748 | 0.005 | 0.255 | 0.742 |
| *Temporal* | 1,160 | 0.083 | 0.781 | 0.003 | 0.086 | 0.778 |
| *Smoothed NDWI_I (NIR-SWIR1/NIR+SWIR1)* | 1060 | 0.217 | 0.392 | 0.001 | 0.219 | 0.391 |
| *Mean* | 290 | 0.065 | 0.457 | 0.005 | 0.070 | 0.453 |
| *Temporal* | 1,160 | -0.055 | 0.497 | 0.003 | -0.052 | 0.494 |
| *Smoothed (Tasseled Cap Greenness)* | 1060 | 0.090 | 0.169 | 0.001 | 0.090 | 0.169 |
| *Mean* | 290 | 0.025 | 0.270 | 0.003 | 0.027 | 0.267 |
| *Temporal* | 1,160 | -0.053 | 0.304 | 0.001 | -0.052 | 0.302 |
| *Smoothed (Tasseled Cap Wetness)* | 1060 | -0.129 | -0.088 | 0.000 | -0.129 | -0.089 |
| *Mean* | 290 | -0.205 | -0.032 | 0.002 | -0.204 | -0.034 |
| *Temporal* | 1,160 | -0.326 | -0.020 | 0.001 | -0.325 | -0.021 |
| *Smoothed (Ground Height)* | 1160 | 26.20 | 940.67 | 6.58 | 32.79 | 934.09 |
| *Mean* | 290 | 26.20 | 940.67 | 13.19 | 39.39 | 927.48 |
| *Static* | 1,160 | 26.20 | 940.67 | 6.58 | 32.79 | 934.09 |
| *Smoothed (Solar Radiation)* | 1160 | 4254.61 | 7709.27 | 20.44 | 4275.05 | 7688.83 |
| *Mean* | 290 | 4254.61 | 7709.27 | 40.93 | 4295.55 | 7668.34 |
| *Temporal* | 1,160 | 4254.61 | 7709.27 | 20.44 | 4275.05 | 7688.83 |
| **04_Wetland** | **Samples** | **Min** | **Max** | **Standard Error** | **Range** | |
| Smoothed (NDVI) | 90 | 0.361 | 0.591 | 0.006 | 0.367 | *0.585* |
| Mean | 26 | -0.056 | 0.643 | 0.029 | -0.026 | *0.614* |
| *Temporal* | 104 | -0.109 | 0.717 | 0.018 | -0.092 | *0.699* |
| *Smoothed (SWIR1-SWIR2/SWIR1+SWIR2)* | 90 | 0.268 | 0.384 | 0.003 | 0.271 | 0.381 |
| *Mean* | 26 | 0.176 | 0.382 | 0.008 | 0.185 | 0.373 |
| *Static* | 104 | 0.166 | 0.460 | 0.006 | 0.172 | 0.455 |
| *Smoothed (Tasseled Cap Greenness)* | 90 | 0.020 | 0.104 | 0.002 | 0.022 | 0.102 |
| *Mean* | 26 | -0.043 | 0.142 | 0.009 | -0.033 | 0.132 |
| *Temporal* | 104 | -0.062 | 0.175 | 0.006 | -0.056 | 0.169 |
| *Smoothed (Tasseled Cap Wetness)* | 90 | -0.150 | -0.035 | 0.003 | -0.146 | -0.038 |
| *Mean* | 26 | -0.205 | 0.015 | 0.013 | -0.193 | 0.002 |
| *Temporal* | 104 | -0.225 | 0.017 | 0.007 | -0.218 | 0.010 |
| *Smoothed (Wetness Index)* | 104 | -0.366 | 7.620 | 0.289 | -0.078 | 7.332 |
| *Mean* | 26 | -0.366 | 7.620 | 0.586 | 0.220 | 7.034 |
| *Temporal* | 104 | -0.366 | 7.620 | 0.289 | -0.078 | 7.332 |
| **05_Built Up** | **Samples** | **Min** | **Max** | **Standard Error** | **Range** | |
| *Smoothed (Slope)* | 1220 | 0.03 | 14.30 | 0.08 | 0.11 | 14.22 |
| *Mean* | 305 | 0.028 | 14.30 | 0.156 | 0.183 | 14.14 |
| *Static* | 1,220 | 0.028 | 14.30 | 0.078 | 0.105 | *14.220* |
| *Smoothed NDVI (NIR-R / NIR+R)* | 1120 | 0.253 | 0.557 | 0.002 | 0.256 | 0.555 |
| *Mean* | 305 | 0.080 | 0.730 | 0.007 | 0.087 | 0.722 |
| *Temporal* | 1,220 | 0.038 | 0.798 | 0.004 | 0.042 | 0.794 |
| *Smoothed (NIR-SWIR2/NIR+SWIR2)* | 1120 | 0.168 | 0.469 | 0.002 | 0.170 | 0.467 |
| *Mean* | 305 | 0.007 | 0.646 | 0.008 | 0.015 | 0.638 |
| *Temporal* | 1,220 | -0.037 | 0.726 | 0.004 | -0.033 | 0.722 |
| *Smoothed (SWIR1-SWIR2/SWIR1+SWIR2)* | 1120 | 0.153 | 0.270 | 0.001 | 0.154 | 0.269 |
| *Mean* | 305 | 0.038 | 0.391 | 0.004 | 0.042 | 0.387 |
| *Temporal* | 1,220 | 0.018 | 0.447 | 0.002 | 0.020 | 0.445 |
| Smoothed (Tasseled Cap Greenness) | 1120 | -0.014 | 0.088 | 0.001 | -0.014 | 0.087 |
| Mean | 305 | -0.149 | 0.175 | 0.003 | -0.147 | 0.173 |

| | | | | | | |
|---|---|---|---|---|---|---|
| *Temporal* | 1,220 | -0.169 | 0.207 | 0.001 | -0.167 | 0.206 |
| *Smoothed NDWI_I (NIR-SWIR1/NIR+SWIR1)* | 1120 | 0.015 | 0.228 | 0.002 | 0.016 | 0.226 |
| *Mean* | 305 | -0.140 | 0.371 | 0.005 | -0.134 | 0.366 |
| *Temporal* | 1,220 | -0.163 | 0.423 | 0.003 | -0.160 | 0.420 |
| **06_Water** | **Samples** | **Min** | **Max** | **Standard Error** | **Range** | |
| Mean NDWI_I (NIR-SWIR1/NIR+SWIR1) | 8 | 0.233 | 0.465 | 0.027 | 0.260 | *0.438* |
| *Temporal* | 32 | -0.024 | 0.523 | 0.029 | 0.005 | *0.494* |
| *Mean (Tasselled Cap Greenness)* | 8 | -0.016 | 0.018 | 0.005 | -0.012 | 0.013 |
| *Temporal* | 32 | -0.046 | 0.055 | 0.003 | -0.043 | 0.051 |
| *Mean GNDVI (NIR-Green/NIR+Green)* | 8 | 0.045 | 0.309 | 0.035 | 0.080 | 0.274 |
| *Temporal* | 32 | -0.318 | 0.519 | 0.040 | -0.277 | 0.479 |
| **07_Cropland** | **Samples** | **Min** | **Max** | **Standard Error** | **Range** | |
| *Smoothed NDVI (NIR-Red / NIR+Red)* | *1404* | 0.254 | 0.744 | 0.003 | 0.257 | 0.740 |
| *Mean* | *376* | 0.176 | 0.735 | 0.006 | 0.182 | 0.729 |
| *Temporal* | 1,504 | -0.078 | 0.833 | 0.005 | -0.073 | *0.828* |
| *Smoothed (BN)* | *1404* | -0.789 | -0.456 | 0.002 | -0.787 | -0.458 |
| *Mean* | *376* | -0.791 | -0.283 | 0.004 | -0.787 | -0.287 |
| *Temporal* | 1,504 | -0.845 | -0.014 | 0.003 | -0.841 | -0.017 |
| *Smoothed (Tasselled Cap Greenness)* | *1404* | -0.004 | 0.153 | 0.001 | -0.003 | 0.152 |
| *Mean* | *376* | -0.032 | 0.195 | 0.002 | -0.030 | 0.193 |
| *Temporal* | 1,504 | -0.067 | 0.249 | 0.002 | -0.066 | 0.247 |
| *Smoothed (Tasseled Cap Wetness)* | *1404* | -0.224 | -0.078 | 0.001 | -0.223 | -0.079 |
| *Mean* | *376* | -0.306 | -0.040 | 0.003 | -0.303 | -0.043 |
| *Temporal* | 1,504 | -0.406 | 0.033 | 0.002 | -0.405 | 0.031 |
| *Smoothed (Ground Height)* | *1504* | 6.07 | 700.61 | 4.31 | 10.38 | 696.30 |
| *Mean* | *376* | 6.07 | 700.61 | 8.63 | 14.70 | 691.97 |
| *Temporal* | 1,504 | 6.070 | 700.61 | 4.31 | 10.38 | 696.30 |
| *Smoothed (Slope)* | *1504* | 0.088 | 21.536 | 0.094 | 0.182 | 21.443 |
| *Mean* | *376* | 0.088 | 21.536 | 0.187 | 0.276 | 21.349 |
| *Temporal* | 1,504 | 0.088 | 21.536 | 0.094 | 0.182 | 21.443 |
| ***NDVI_Cultivation/Harvest_First Quartile*** | **1,081** | | | **$R2 = 0.93$** | **0.133** | **0.321** |
| *B2 (Blue)* | *1,081* | | | $R2 = 0.84$ | 0.042 | 0.058 |
| *B8 (NIR)* | *1,081* | | | $R2 = 0.76$ | 0.222 | 0.240 |
| *B11 (SWIR1)* | *1,081* | | | $R2 = 0.86$ | 0.147 | 0.178 |
| ***NDVI_Senescence_Third Quartile*** | ***1,081*** | | | **$R2 = 0.93$** | **0.321** | **0.563** |
| *B2 (Blue)* | *1,081* | | | $R2 = 0.84$ | 0.058 | 0.074 |
| *B8 (NIR)* | *1,081* | | | $R2 = 0.76$ | 0.240 | 0.273 |
| *B11 (SWIR1)* | *1,081* | | | $R2 = 0.86$ | 0.178 | 0.252 |
| ***NDVI_Peak growing_Last Quartile*** | ***1,081*** | | | **$R2 = 0.93$** | **0.563** | **0.780** |
| *B2 (Blue)* | *1,081* | | | $R2 = 0.84$ | 0.074 | 0.109 |
| *B8 (NIR)* | *1,081* | | | $R2 = 0.76$ | 0.273 | 0.294 |
| *B11 (SWIR1)* | *1,081* | | | $R2 = 0.86$ | 0.252 | 0.337 |

Table S2. Smoothed, mean, and raw metric performances. Thematic agreement variations can be seen for each metric set.

| Metric Performances | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **01_Dense Short Vegetation** | **Samples** | **Level 0** | **Level 1** | **Level 2** | **Level 3** | **Thematic Agreement** | **Height Agreement** | **Structural Agreement** |
| *Smoothed Metrics* | 36,728 | 1,561 | 8,981 | 16,991 | 9,195 | 25,600 | 19,364 | 25,584 |
| | | 4.3% | 24.5% | 46.3% | 25.0% | 69.7% | 52.7% | 69.7% |
| *Mean Metrics* | *36,728* | 1,505 | 8,737 | 17,001 | 9,485 | 26,246 | 19,364 | 25,584 |
| | | 4.1% | 23.8% | 46.3% | 25.8% | 71.5% | 52.7% | 69.7% |
| *Temporal Raw Metrics* | *36,728* | 110 | 5,715 | 17,572 | 13,331 | 35,904 | 19,364 | 25,584 |
| | | 0.3% | 15.6% | 47.8% | 36.3% | 97.8% | 52.7% | 69.7% |
| **02_Open Tree Cover** | **Samples** | **Level 0** | **Level 1** | **Level 2** | **Level 3** | **Thematic Agreement** | **Height Agreement** | **Structural Agreement** |
| *Smoothed Metrics* | 36,788 | 3,459 | 11,078 | 15,974 | 6,277 | 15,333 | 22,582 | 23,942 |
| | | 9.4% | 30.1% | 43.4% | 17.1% | 41.7% | 61.4% | 65.1% |
| *Mean Metrics* | *36,788* | 1,967 | 9,332 | 15,891 | 9,598 | 25,987 | 19,979 | 23,942 |
| | | 5.3% | 25.4% | 43.2% | 26.1% | 70.6% | 54.3% | 65.1% |
| *Temporal Raw Metrics* | *36,788* | 912 | 7,802 | 16,442 | 11,632 | 31,661 | 19,979 | 23,942 |
| | | 2.5% | 21.2% | 44.7% | 31.6% | 86.1% | 54.3% | 65.1% |
| **03_Dense Tree Cover** | **Samples** | **Level 0** | **Level 1** | **Level 2** | **Level 3** | **Thematic Agreement** | **Height Agreement** | **Structural Agreement** |
| *Smoothed Metrics* | 28,397 | 2,005 | 14,733 | 10,301 | 1,358 | 3,669 | 21,221 | 14,519 |
| | | 7.1% | 51.9% | 36.3% | 4.8% | 12.9% | 74.7% | 51.1% |
| *Mean Metrics* | *28,397* | 751 | 8,659 | 12,815 | 6,172 | 17,065 | 21,221 | 14,519 |
| | | 2.6% | 30.5% | 45.1% | 21.7% | *60.1%* | 74.7% | 51.1% |
| *Temporal Raw Metrics* | *28,397* | 450 | 7,090 | 13,503 | 7,354 | 20,418 | 21,221 | 14,519 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | 1.6% | 25.0% | 47.6% | 25.9% | *71.9%* | 74.7% | 51.1% |
| **04_Wetland Cover** | **Samples** | **Level 0** | **Level 1** | **Level 2** | **Level 3** | **Thematic Agreement** | **Height Agreement** | **Structural Agreement** |
| *Smoothed Metrics* | 1,245 | 162 | 252 | 739 | 92 | 134 | 1,000 | 872 |
| | | 13.0% | 20.2% | 59.4% | 7.4% | 10.8% | 80.3% | 70.0% |
| *Mean Metrics* | *1,245* | 151 | 220 | 611 | 263 | *359* | 1,000 | 872 |
| | | 12.1% | 17.7% | 49.1% | 21.1% | *28.8%* | 80.3% | 70.0% |
| *Temporal Raw Metrics* | *1,245* | 103 | 212 | 492 | 438 | *638* | 1,000 | 872 |
| | | 8.3% | 17.0% | 39.5% | 35.2% | 51.2% | 80.3% | 70.0% |
| **05_Built Up** | **Samples** | **Level 0** | **Level 1** | **Level 2** | **Level 3** | **Thematic Agreement** | **Height Agreement** | **Structural Agreement** |
| *Smoothed Metrics* | 24,602 | 2,722 | 5,316 | 12,506 | 4,058 | 6,742 | 16,618 | 19,142 |
| | | 11.1% | 21.6% | 50.8% | 16.5% | 27.4% | 67.5% | 77.8% |
| *Mean Metrics* | *24,602* | 986 | 4,006 | 8,430 | 11,180 | *18,646* | 16,618 | 19,142 |
| | | 4.0% | 16.3% | 34.3% | 45.4% | *75.8%* | 67.5% | 77.8% |
| *Temporal Raw Metrics* | *24,602* | 424 | 3,627 | 7,450 | 13,101 | 22,070 | 16,618 | 19,142 |
| | | 1.7% | 14.7% | 30.3% | 53.3% | 89.7% | 67.5% | 77.8% |
| **06_Water** | **Samples** | **Level 0** | **Level 1** | **Level 2** | **Level 3** | **Thematic Agreement** | **Height Agreement** | **Structural Agreement** |
| *Smoothed Metrics* | 1,864 | - | 49 | 1,132 | 683 | 754 | 1,860 | 1,748 |
| | | 0.0% | 2.6% | 60.7% | 36.6% | 40.5% | 99.8% | 93.8% |
| *Mean Metrics* | *1,864* | 2 | 110 | 1,732 | 20 | *26* | *1,860* | 1,748 |
| | | 0.1% | 5.9% | 92.9% | 1.1% | *1.4%* | *99.8%* | 93.8% |
| *Temporal Raw Metrics* | *1,864* | 0 | 49 | 1,132 | 683 | 754 | 1,860 | 1,748 |
| | | 0.0% | 2.6% | 60.7% | 36.6% | 40.5% | 99.8% | 93.8% |
| **07_Cropland** | **Samples** | **Level 0** | **Level 1** | **Level 2** | **Level 3** | **Thematic Agreement** | **Height Agreement** | **Structural Agreement** |
| *Smoothed Metrics* | 38,181 | 4,747 | 8,725 | 17,219 | 7,490 | 13,620 | 23,741 | 28,272 |
| | | 12.4% | 22.9% | 45.1% | 19.6% | 35.7% | 62.2% | 74.0% |
| *Cultivation/Harvest* | 8,216 | 1,199 | 807 | 5,576 | 634 | 952 | 6,149 | 6,760 |
| | | 7.6% | 5.1% | 35.4% | 4.0% | 6.0% | 39.0% | 42.9% |
| *Senescence* | 14,192 | 1,202 | 3,443 | 5,302 | 4,245 | 7,539 | 8,781 | 10,462 |
| | | 8.5% | 24.3% | 37.4% | 29.9% | 53.1% | 61.9% | 73.7% |
| *Peak Growing* | 15,773 | 2,346 | 4,475 | 6,341 | 2,611 | 5,129 | 8,811 | 11,050 |
| | | 28.6% | 54.5% | 77.2% | 31.8% | 62.4% | 107.2% | 134.5% |
| *Mean Metrics* | *38,181* | 3,258 | 9,209 | 14,403 | 11,311 | 19,935 | 23,741 | 28,272 |
| | | 8.5% | 24.1% | 37.7% | 29.6% | 52.2% | 62.2% | 74.0% |
| *Cultivation/Harvest* | *8,216* | 633 | 1,129 | 4,103 | 2,351 | 3,479 | 6,149 | 6,760 |
| | | 4.0% | 7.2% | 26.0% | 14.9% | 22.1% | 39.0% | 42.9% |
| *Senescence* | *14,192* | 923 | 3,483 | *4,742* | *5,044* | 8,856 | *8,781* | *10,462* |
| | | 6.5% | 24.5% | 33.4% | 35.5% | 62.4% | 61.9% | 73.7% |
| *Peak Growing* | *15,773* | 1,702 | 4,597 | 5,558 | 3,916 | 7,600 | 8,811 | 11,050 |
| | | 20.7% | 56.0% | 67.6% | 47.7% | 92.5% | 107.2% | 134.5% |
| *Raw Temporal Metrics* | *38,181* | 2,005 | 9,558 | 11,774 | 14,844 | 25,625 | 23,741 | 28,272 |
| | | 5.3% | 25.0% | 30.8% | 38.9% | 67.1% | 62.2% | 74.0% |
| *Cultivation/Harvest* | *8,216* | 248 | 1,355 | 2,501 | 4,112 | 5,784 | 6,149 | 6,760 |
| | | 1.6% | 8.6% | 15.9% | 26.1% | 36.7% | 39.0% | 42.9% |
| *Senescence* | *14,192* | 688 | 3,590 | 4,221 | 5,693 | 9,868 | 8,781 | 10,462 |
| | | 4.8% | 25.3% | 29.7% | 40.1% | 69.5% | 61.9% | 73.7% |
| *Peak Growing* | *15,773* | 1,069 | 4,613 | 5,052 | 5,039 | 9,973 | 8,811 | 11,050 |
| | | 13.0% | 56.1% | 61.5% | 61.3% | 121.4% | 107.2% | 134.5% |

Table S3. Training sample classification, refinement, and filtering.

| Classes | | 1. Classification | | | | | 2. Refinement | | 3. Filtering | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Sample (-in) | Level-0 | Level-1 | Level-2 | Level-3 | Samples (-out) | Level-2 | Level-3 | Level-0 | Level-1 | Level-2 |
| 1. Dense Short Vegetation | 36,728 | 110 | 5,715 | 17,572 | 13,331 | 36,728 | 17,172 | 13,331 | 110 | 5,715 | 400 |
| 2. Open Tree Cover | 36,788 | 912 | 7,802 | 16,442 | 11,632 | 36,788 | 5,413 | 11,632 | 912 | 7,802 | 11,029 |
| 3. Dense Tree Cover | 28,397 | 450 | 7,090 | 13,503 | 7,354 | 28,397 | 7,279 | 7,354 | 450 | 7,090 | 6,224 |
| 4. Wetland | 1,245 | 103 | 212 | 492 | 438 | 1,245 | 492 | 438 | 103 | 212 | 0 |
| 5. Built Up | 24,602 | 424 | 3,627 | 7,450 | 13,101 | 24,602 | 5,887 | 13,101 | 424 | 3,627 | 1,563 |
| 6. Water | 1,864 | 0 | 49 | 1,132 | 683 | 1,864 | 1,132 | 683 | 0 | - | 0 |
| 7.Cropland | 38,181 | 2,005 | 9,558 | 11,774 | 14,844 | 0 | - | - | 0 | 49 | 0 |
| 7.Peak Growing | 0 | 1,069 | 4,613 | 5,052 | 5,039 | 15,773 | 2,203 | 5,039 | 1,069 | 4,613 | 2,849 |
| 8. Senescence | 0 | 688 | 3,590 | 4,221 | 5,693 | 14,192 | 1,606 | 5,693 | 688 | 3,590 | 2,615 |
| 9. Cultivation | 0 | 248 | 1,355 | 2,501 | 4,112 | 8,216 | 536 | 4,112 | 248 | 1,355 | 1,965 |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Total** | **167,805** | **4,004** | **34,053** | **68,365** | **61,383** | **167,805** | **41,720** | **61,383** | **4,004** | **34,053** | **26,645** |
| | **100%** | **2%** | **20%** | **41%** | **37%** | **100%** | **25%** | **37%** | **2%** | **20%** | **16%** |

Table S4. Permutation-based feature importance’s. First, all features were tested. Then, 10 feature combinations were identified. Overfitting features are shaded and correlated features are in bold. Note that band correlations were not considered and default parameters were used for 20% unseen test data.

*Overfitting and Correlation Evaluations*

| | Features | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** | **12** | **13** | **14** | **15** | **16** | **17** | **18** | **19** | **20** | **21** | **22** | **23** | **24** | **25** | **Total** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Training Dataset | *25* | SWIR 1 2 | NDWI I | B11 | RS2 | MCARI | Tass Wet | B3 | B2 | B1 | BG | NIR SWIR 2 | NDWI II | NDVI | B12 | BG | BN | B4 | B5 | Tass GN | IRECI | B6 | B8A | B7 | B8 | SAVI | |
| | *Overfitting* | 0.11 | 0.035 | 0.032 | 0.032 | 0.031 | 0.03 | 0.023 | 0.022 | 0.02 | 0.02 | 0.02 | 0.019 | 0.018 | 0.018 | 0.017 | 0.016 | 0.015 | 0.012 | 0.012 | 0.012 | 0.01 | 0.008 | 0.007 | 0.006 | 0.005 | 0.550 |
| | 2 | +/- 0.001 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.001 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.002 |
| Test Data | 25 | SWIR 1 2 | RS2 | NDWI I | B11 | MCARI | Tass Wet | B3 | B2 | BG | B1 | NIR SWIR 2 | NDWI II | B12 | BG | BN | NDVI | Tass GN | B4 | IRECI | B5 | B6 | B8A | B7 | SAVI | B8 | |
| | *Correlations* | 0.108 | 0.035 | 0.033 | 0.033 | 0.032 | 0.031 | 0.025 | 0.024 | 0.022 | 0.02 | 0.02 | 0.019 | 0.019 | 0.018 | 0.017 | 0.016 | 0.015 | 0.014 | 0.013 | 0.013 | 0.012 | 0.09 | 0.09 | 0.08 | 0.08 | 0.879 |
| | *0* | +/- 0.001 | +/- 0.001 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.001 | +/- 0.000 | +/- 0.001 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.000 | +/- 0.004 |

*Feature Permutations*

| | Features | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** | **12** | **13** | **14** | **15** | **Total** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Model 1** | 15 | SWIR_1_2 | B11 | RS2 | NDWI_II | B1 | B3 | BG | NDVI | B2 | B5 | B4 | TassGN | B6 | SAVI | B8A | |
| | *Correlations* | 0.135 | 0.097 | 0.027 | 0.023 | 0.021 | 0.021 | **0.018** | **0.018** | 0.017 | 0.015 | 0.01 | 0.008 | 0.008 | 0.006 | 0.006 | 0.43 |
| | 1 | +/- 0.002 | +/- 0.003 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.000 | +/- 0.017 |
| **Model 2** | 14 | SWIR_1_2 | B11 | TassWet | B3 | RS2 | NDWI_II | B2 | NDVI | B1 | NIR_SWIR_2 | B5 | B4 | B6 | B8A | | |
| | *Correlations* | 0.126 | 0.029 | 0.027 | 0.025 | 0.018 | 0.016 | 0.015 | 0.015 | 0.013 | 0.012 | 0.011 | 0.08 | 0.07 | 0.002 | | 0.459 |
| | 0 | +/- 0.002 | +/- 0.001 | +/- 0.001 | +/- 0.002 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | | +/- 0.016 |
| **Model 3** | 14 | SWIR_1_2 | B3 | B11 | TassWet | NDVI | RS2 | NIR_SWIR_2 | B2 | B5 | B1 | BN | B4 | B6 | B8A | | |
| | *Correlations* | 0.125 | 0.029 | 0.026 | 0.023 | 0.019 | 0.017 | 0.012 | 0.011 | 0.011 | 0.009 | 0.006 | 0.005 | 0.002 | 0.002 | | 0.297 |
| | 0 | +/- 0.002 | +/- 0.001 | +/- 0.002 | +/- 0.002 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.002 | +/- 0.001 | +/- 0.001 | | +/- 0.018 |
| **Model 4** | 15 | SWIR_1_2 | B11 | RS2 | B3 | NIR_SWIR_2 | NDWI_II | IRECI | NDVI | B1 | B2 | BN | B5 | B4 | B6 | B8A | |
| | *Correlations* | 0.127 | 0.046 | 0.027 | 0.021 | 0.019 | **0.015** | **0.015** | 0.014 | 0.012 | 0.012 | 0.01 | 0.009 | 0.007 | 0.006 | 0.003 | 0.343 |
| | 1 | +/- 0.001 | +/- 0.002 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.015 |
| **Model 5** | 15 | SWIR_1_2 | B11 | TassWet | B3 | RS2 | NDVI | NIR_SWIR_2 | B2 | IRECI | B1 | BN | B4 | B5 | B6 | B8A | |
| | *Correlations* | 0.118 | 0.025 | 0.024 | 0.024 | 0.019 | 0.017 | 0.014 | 0.01 | **0.008** | 0.008 | **0.008** | 0.006 | 0.005 | 0.003 | 0.003 | 0.292 |
| | 1 | +/- 0.002 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.002 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.017 |
| **Model 6** | 15 | SWIR_1_2 | NDWI_I | NDVI | B3 | TassWet | B11 | RS2 | GB | B2 | B5 | B1 | BN | B4 | B6 | B8A | |
| | *Correlations* | 0.12 | 0.03 | 0.022 | 0.021 | 0.019 | 0.019 | 0.018 | 0.015 | 0.01 | 0.09 | 0.08 | 0.06 | 0.06 | 0.03 | 0.004 | 0.598 |
| | 0 | +/- 0.002 | +/- 0.002 | +/- 0.002 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.000 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.017 |
| **Model 7** | 15 | SWIR_1_2 | B11 | RS2 | B3 | NIR_SWIR_2 | GB | IRECI | B2 | B1 | B5 | B4 | NDVI | B6 | SAVI | B8A | |
| | *Correlations* | 0.121 | 0.036 | 0.024 | 0.021 | 0.018 | 0.017 | 0.01 | 0.01 | 0.08 | 0.07 | 0.05 | 0.04 | 0.03 | 0.01 | 0.01 | 0.547 |
| | 0 | +/- 0.002 | +/- 0.001 | +/- 0.002 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.002 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.018 |
| **Model 8** | 15 | SWIR_1_2 | NDWI_I | B11 | B3 | RS2 | B2 | BN | B1 | NDVI | IRECI | B5 | TassWT | B4 | B6 | B8A | |
| | *Correlations* | 0.113 | 0.04 | 0.029 | 0.026 | 0.025 | 0.012 | 0.011 | 0.01 | 0.01 | 0.009 | 0.008 | 0.007 | 0.007 | 0.005 | 0.004 | 0.316 |
| | 0 | +/- 0.001 | +/- 0.002 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.002 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.017 |
| **Model 9** | 15 | SWIR_1_2 | B11 | RS2 | B3 | NIR_SWIR_2 | BG | IRECI | B2 | NDVI | BN | B5 | B1 | B4 | B6 | B8A | |
| | *Correlations* | 0.12 | 0.037 | 0.025 | 0.02 | 0.019 | 0.017 | 0.01 | 0.009 | **0.007** | **0.007** | 0.006 | 0.006 | 0.004 | 0.004 | 0.003 | 0.294 |
| | 1 | +/- 0.002 | +/- 0.002 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.002 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.018 |
| **Model 10** | 14 | SWIR_1_2 | B11 | RS2 | B3 | NIR_SWIR_2 | BG | NDVI | B2 | B5 | B1 | B4 | BN | B6 | B8A | | |
| | *Correlations* | 0.121 | 0.036 | 0.023 | 0.021 | 0.02 | 0.018 | 0.011 | 0.009 | 0.009 | 0.007 | 0.005 | 0.005 | 0.004 | 0.003 | | 0.292 |
| | 0 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.002 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | +/- 0.001 | | +/- 0.015 |

Table S5. Supervised mean-weighed sample design. The upper table provides class weights from a preliminary classification using model 10. The lower table denotes mean-weighed allocations.

| Classes | PixelSum | Percentage % | Area [metre^2] | Class Weight |
|---|---|---|---|---|
| 1 | 7392461 | 6.788441372 | 6677624451 | 0.06788 |
| 2 | 12937193 | 11.88012709 | 11686191690 | 0.1188 |
| 3 | 17522660 | 16.09092697 | 15828252982 | 0.16091 |
| 4 | 1314928 | 1.207488499 | 1187777029 | 0.01207 |
| 5 | 11404567 | 10.47272815 | 10301767632 | 0.10473 |
| 6 | 25923388 | 23.80525235 | 23416646982 | 0.23805 |
| 7 | 13605844 | 12.49414428 | 12290185443 | 0.12494 |
| 8 | 15131535 | 13.89517486 | 13668345102 | 0.13895 |
| 9 | 3665190 | 3.365716428 | 3310773281 | 0.03366 |

ICESat-2 Ground Track ATL0820190904

| Classes | Weighed | Equal | Mean-Weighed | L3 reference samples available |
|---|---|---|---|---|
| 1 | 20 | 33 | 27 | 19 |
| 2 | 35 | 33 | 34 | 53 |
| 3 | 47 | 33 | 40 | 42 |
| 4 | 4 | 33 | 18 | 2 |
| 5 | 31 | 33 | 32 | 39 |
| 6 | 70 | 33 | 52 | 10 |
| 7 | 37 | 33 | 35 | 35 |
| 8 | 41 | 33 | 37 | 57 |
| 9 | 10 | 33 | 21 | 12 |
| Total | 295 | 295 | 295 | 269 |

Table S6. Mean absolute difference estimations. Values represent per-class absolute differences to estimate the mean difference for each model and the net average of model estimates.

| Mean Absolute Difference | | *Dense Short Vegetation* | *Open Tree Cover* | *Dense Tree Cover* | *Wetland* | *Built Up* | *Water* | *Cropland Strata* | *Mean Difference* | *OA* |
|---|---|---|---|---|---|---|---|---|---|---|
| Model 1 | SMW | 37.5% | 66.7% | 75.0% | - | 77.3% | 80.0% | 80.8% | | 73.9% |
| | SRS | 36.7% | 42.6% | 66.7% | - | 84.4% | 84.6% | 78.9% | | 64.9% |
| | Absolute Difference | 0.8% | 24.0% | 8.3% | - | 7.1% | 4.6% | 1.9% | 7.80% | 9.0% |
| Model 2 | SMW | 75.0% | 61.4% | 82.5% | - | 90.9% | 80.0% | 74.0% | | 74.6% |
| | SRS | 43.3% | 37.7% | 71.8% | - | 87.5% | 84.6% | 73.3% | | 63.8% |
| | Absolute Difference | 31.7% | 23.7% | 10.7% | - | 3.4% | 4.6% | 0.7% | 12.47% | 10.8% |
| Model 3 | SMW | 75.0% | 63.2% | 80.0% | - | 86.4% | 80.0% | 76.9% | | 74.6% |
| | SRS | 50.0% | 41.0% | 71.8% | - | 84.4% | 84.6% | 75.6% | | 65.7% |
| | Absolute Difference | 25.0% | 22.2% | 8.2% | - | 2.0% | 4.6% | 1.4% | 10.56% | 9.0% |
| Model 4 | SMW | 75.0% | 59.7% | 75.0% | - | 81.8% | 80.0% | 72.1% | | 70.5% |
| | SRS | 50.0% | 36.1% | 64.1% | - | 81.3% | 84.6% | 74.4% | | 62.7% |
| | Absolute Difference | 25.0% | 23.6% | 10.9% | - | 0.6% | 4.6% | 2.3% | 11.17% | 7.8% |
| Model 5 | SMW | 50.0% | 64.9% | 80.0% | - | 77.3% | 80.0% | 76.9% | | 72.8% |
| | SRS | 36.7% | 44.3% | 71.8% | - | 84.4% | 84.6% | 78.9% | | 66.0% |
| | Absolute Difference | 13.3% | 20.7% | 8.2% | - | 7.1% | 4.6% | 2.0% | 9.31% | 6.7% |
| Model 6 | SMW | 75.0% | 63.2% | 80.0% | - | 81.8% | 80.0% | 77.9% | | 75.0% |
| | SRS | 33.3% | 45.9% | 71.8% | - | 84.4% | 84.6% | 77.8% | | 65.7% |
| | Absolute Difference | 41.7% | 17.3% | 8.2% | - | 2.6% | 4.6% | 0.1% | 12.40% | 9.3% |
| Model 7 | SMW | 50.0% | 64.9% | 77.5% | - | 84.1% | 80.0% | 79.8% | | 75.0% |
| | SRS | 40.0% | 44.3% | 69.2% | - | 84.4% | 84.6% | 78.9% | | 66.0% |
| | Absolute Difference | 10.0% | 20.7% | 8.3% | - | 0.3% | 4.6% | 0.9% | 7.46% | 9.0% |
| Model 8 | SMW | 75.0% | 64.9% | 82.5% | - | 79.5% | 80.0% | 79.8% | | 75.4% |
| | SRS | 40.0% | 42.6% | 69.2% | - | 81.3% | 84.6% | 80.0% | | 65.7% |
| | Absolute Difference | 35.0% | 22.3% | 13.3% | - | 1.7% | 4.6% | 0.2% | 12.85% | 9.7% |
| Model 9 | SMW | 50.0% | 64.9% | 77.5% | - | 77.3% | 80.0% | 80.8% | | 74.3% |
| | SRS | 36.7% | 42.6% | 69.2% | - | 87.5% | 84.6% | 78.9% | | 65.7% |
| | Absolute Difference | 13.3% | 22.3% | 8.3% | - | 10.2% | 4.6% | 1.9% | 10.11% | 8.6% |
| Model 10 | SMW | 50.0% | 75.4% | 77.5% | - | 77.3% | 80.0% | 80.8% | | 76.9% |
| | SRS | 30.0% | 49.2% | 69.2% | - | 84.4% | 84.6% | 78.9% | | 66.0% |
| | Absolute Difference | 20.0% | 26.3% | 8.3% | - | 7.1% | 4.6% | 1.9% | 11.35% | 10.8% |
| MAD | | | | | | | | | 10.55% | 9.07% |

Note: Pixel-based producer's and overall accuracies (OA) are estimated for the Mean Absolute Difference (MAD). Wetland areas were not included given the lack of unit samples.

Table S7. Mean total error estimations. Filters are evaluated to determine which performed best. Structural agreements consider height correlations and thematic agreements. Height agreement consider the height thresholds and thematic agreements. Lastly, SRS considers photo-inspected samples and SMW considers photo-inspected samples with structural and height agreements. Note the 33% difference between SRS and SMW.

| | L2: Structural Agreement | L2: Height Agreement | SRS | SMW |
|---|---|---|---|---|
| Model 1 | 0.743 | 0.556 | 0.397 | 0.287 |
| Model 2 | 0.575 | 0.515 | 0.406 | 0.248 |
| Model 3 | 0.53 | 0.509 | 0.424 | 0.237 |
| Model 4 | 0.721 | 0.501 | 0.46 | 0.283 |
| Model 5 | 0.696 | 0.559 | 0.344 | 0.234 |
| Model 6 | 0.553 | 0.484 | 0.398 | 0.291 |
| Model 7 | 0.659 | 0.591 | 0.37 | 0.279 |
| Model 8 | 0.601 | 0.562 | 0.351 | 0.273 |
| Model 9 | 0.672 | 0.543 | 0.399 | 0.246 |
| Model 10 | 0.671 | 0.571 | 0.369 | 0.243 |

| Total Mean | 0.64 | 0.54 | 0.39 (1.00) | 0.26 (0.67) |
|---|---|---|---|---|

Table S8. Confusion matrix of preliminary land cover classification with a mean-weighed sampling design estimated with area proportions and shown in percentages with 95th percentile confidence intervals. The overall accuracy (OA) and user's accuracy variances were compared with Table S9.

| Reference | Dense Short Vegetation | Open Tree Cover | Dense Tree Cover | Wetland | Built Up | Water | Cropland Peak Growth | Cropland Senescence | Cropland Cultivation | User's Accuracy | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | |
| 1 | 1.07% | 2.86% | 0.72% | 0.00% | 0.36% | 0.00% | 0.72% | 1.07% | 0.00% | 15.79% | ± 16.4% |
| 2 | 0.00% | 9.64% | 0.67% | 0.00% | 0.45% | 0.00% | 0.22% | 0.90% | 0.00% | 81.13% | ± 10.5% |
| 3 | 0.38% | 1.15% | 12.26% | 0.00% | 0.00% | 0.00% | 1.53% | 0.77% | 0.00% | 76.19% | ± 12.9% |
| 4 | 0.00% | 0.00% | 0.00% | 1.21% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 100.00% | ± 0% |
| 5 | 0.00% | 0.00% | 0.27% | 0.00% | 9.13% | 0.00% | 0.00% | 0.81% | 0.27% | 87.18% | ± 10.5% |
| 6 | 0.00% | 0.00% | 0.00% | 2.65% | 0.00% | 21.16% | 0.00% | 0.00% | 0.00% | 88.89% | ± 20.5% |
| 7 | 0.36% | 1.07% | 1.07% | 0.36% | 0.00% | 0.00% | 9.64% | 0.00% | 0.00% | 77.14% | ± 13.9% |
| 8 | 0.49% | 0.24% | 0.00% | 0.00% | 1.71% | 0.00% | 0.24% | 11.21% | 0.00% | 80.70% | ± 10.2% |
| 9 | 0.00% | 0.00% | 0.00% | 0.28% | 0.00% | 0.28% | 0.00% | 0.00% | 2.81% | 83.33% | ± 21.1% |
| Producer's Accuracy | 46.61% ± 23.9% | 64.43% ± 6.9% | 81.81% ± 11.1% | 26.89% ± 0.0% | 78.42% ± 7.6% | 98.69% ± 21.5% | 78.02% ± 11.3% | 76.00% ± 8.0% | 91.26% ± 21.9% | OA = 78.12% ± 6.1% | |

Table S9**.** Confusion matrix of preliminary land cover classification with a stratified random sampling design estimated with area proportions and shown in percentages with 95th percentile confidence intervals.

| Reference | Dense Short Vegetation | Open Tree Cover | Dense Tree Cover | Wetland | Built Up | Water | Cropland Peak Growth | Cropland Senescence | Cropland Cultivation | Users's Accuracy | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | |
| 1 | 2.78% | 2.47% | 0.31% | 0.00% | 0.31% | 0.00% | 0.00% | 0.62% | 0.31% | 40.91% | ± 20.5% |
| 2 | 2.16% | 8.10% | 0.54% | 0.00% | 0.00% | 0.00% | 0.27% | 0.81% | 0.00% | 68.18% | ± 13.8% |
| 3 | 1.46% | 4.39% | 9.87% | 0.00% | 0.00% | 0.00% | 0.37% | 0.00% | 0.00% | 61.36% | ± 14.4% |
| 4 | 0.13% | 0.00% | 0.00% | 0.27% | 0.27% | 0.13% | 0.13% | 0.13% | 0.13% | 22.22% | ± 27.2% |
| 5 | 0.47% | 0.93% | 0.93% | 0.00% | 6.28% | 0.00% | 0.47% | 0.70% | 0.70% | 60.00% | ± 14.3% |
| 6 | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 23.81% | 0.00% | 0.00% | 0.00% | 100.00% | ± 0.0% |
| 7 | 0.89% | 1.79% | 1.79% | 0.00% | 0.00% | 0.00% | 8.03% | 0.00% | 0.00% | 64.29% | ± 17.7% |
| 8 | 1.11% | 0.83% | 0.28% | 0.28% | 0.56% | 0.00% | 0.00% | 10.84% | 0.00% | 78.00% | ± 11.5% |
| 9 | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.22% | 0.00% | 0.00% | 3.14% | 93.33% | ± 12.6% |
| Producer's Accuracy | 30.84% ± 8.4% | 43.77% ± 6.9% | 71.99% ± 10.3% | 49.12% ± 57.3% | 84.73% ± 10.1% | 98.52% ± 18.3% | 86.67% ± 12.7% | 82.75% ± 8.7% | 73.36% ± 8.0% | OA = 73.12% ± 4.5% | |

Table S10. Confusion matrix of informed land cover classification with a mean-weighed sampling design estimated with area proportions and shown in percentages with 95th percentile confidence intervals. The overall accuracy (OA) and the 95% confidence interval margin are compared with Table S8.

| Reference | Dense Short Vegetation | Open Tree Cover | Dense Tree Cover | Wetland | Built Up | Water | Cropland Peak Growth | Cropland Senescence | Cropland Cultivation | User's Accuracy | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | |
| 1 | 1.20% | 0.72% | 0.24% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 55.56% | ± 32.5% |
| 2 | 0.00% | 10.42% | 0.93% | 0.00% | 0.23% | 0.00% | 0.46% | 0.46% | 0.00% | 83.33% | ± 9.9% |
| 3 | 0.40% | 0.79% | 13.11% | 0.00% | 0.40% | 0.00% | 1.99% | 0.00% | 0.00% | 78.57% | ± 12.4% |
| 4 | 0.00% | 0.00% | 0.00% | 1.48% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 100.00% | ± 0 |
| 5 | 0.00% | 0.25% | 0.00% | 0.00% | 9.48% | 0.25% | 0.00% | 0.75% | 0.25% | 86.36% | ± 10.1% |
| 6 | 0.00% | 0.00% | 0.00% | 4.82% | 0.00% | 19.26% | 0.00% | 0.00% | 0.00% | 80.00% | ± 24.8% |
| 7 | 0.00% | 1.52% | 1.14% | 0.38% | 0.00% | 0.00% | 10.24% | 0.00% | 0.00% | 77.14% | ± 13.9% |
| 8 | 0.50% | 0.50% | 0.00% | 0.00% | 0.99% | 0.00% | 0.00% | 13.61% | 0.00% | 87.30% | ± 8.2% |
| 9 | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 3.26% | 100.00% | ± 0.0% |
| Producer's Accuracy | 57.39% ± 22.7% | 73.40% ± 7.3% | 85.04% ± 11.1% | 22.20% ± 0.0% | 85.41% ± 8.9% | 98.72% ± 28.7% | 80.71% ± 11.9% | 91.83% ± 8.0% | 92.90% ± 0.8% | OA = 82.05% ± 6.9% | |

Table S11. 2016-2021 land transitions expressed in percentage of area over total change. Cumulative hectares of change are in parenthesis sorted from rural to semi-urban to depict differences if any.

| | Srunen (rural) (1) | Gading (rural) (2) | Banjarsari (rural) (3) | Gondang (semi-urban) (4) | Jetis Sumur (semi-urban) (5) | Pagerjurang (semi-urban) (6) | Karang Kendal (semi-urban) (7) | Not Affected (rural) (8) | Not Affected (semi-urban) (9) |
|---|---|---|---|---|---|---|---|---|---|
| *Forest > Crop* | 18% (16.4) | 17% (12.3) | 19% (14.5) | 13% (8.1) | 23% (23.4) | 11% (8.5) | 10% (10.9) | 18% ( 19.5) | 14% (20.9) |
| *Forest > Built* | 34% (30.8) | 24% (17.3) | 28% (20.7) | 30% (18.7) | 27% (25.6) | 35% (24.7) | 38% (42.0) | 30% (30.5) | 34% (45.5) |
| *Crop > Built* | 7% (6.6) | 9% (7.1) | 5% (3.8) | 6% (3.5) | 6% (4.7) | 7% (4.7) | 7% (7.2) | 5% (5.0) | 5% (5.8) |
| *Crop > Forest* | 15% (15.0) | 17% (14.1) | 22% (15.4) | 18% (12.3) | 17% (15.4) | 16% (12.7) | 11% (11.4) | 18% (19.5) | 16% (22.4) |
| *Built > Crop* | 9% (9.6) | 13% (10.4) | 9% (6.6) | 11% (7.8) | 7% (6.6) | 8% (7.0) | 5% (5.8) | 8% (9.0) | 6% (8.6) |
| *Built > Forest* | 17% (19.6) | 20% (16.0) | 17% (12.0) | 22% (14.3) | 20% (19.7) | 24% (20.2) | 30% (30.9) | 20% (22.1) | 25% (33.4) |
| *Total (Ha)* | 97.9 | 77.2 | 73 | 64.8 | 95.5 | 77.8 | 108.1 | 105.7 | 136.5 |

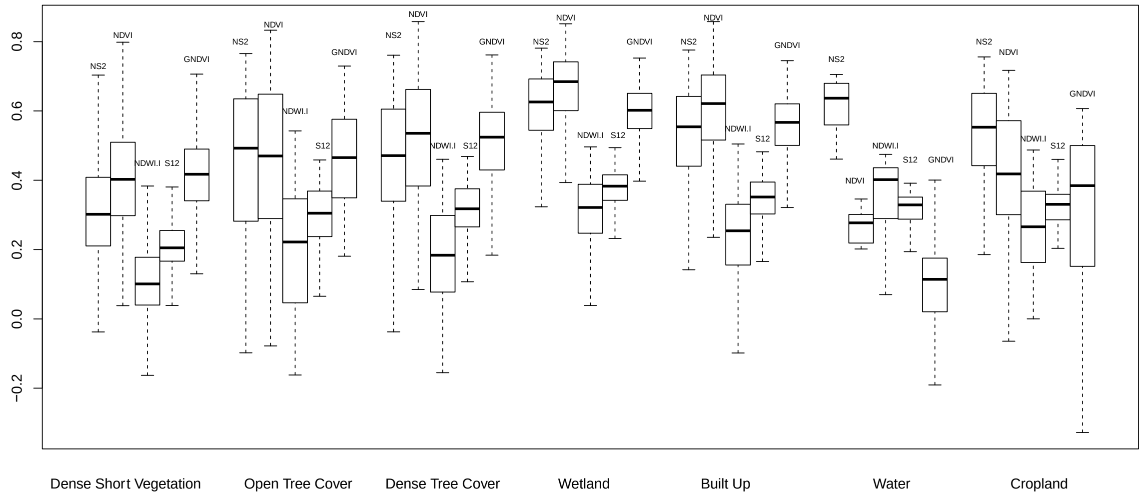



Figure S2. Box plot distributions of spectral amplitude for static and multitemporal metric units. Whiskers are located at 25th, median, and 75th quartiles. Inter-class comparisons highlight meaningful differences between predictors (e.g., BN for water and slope for built-up).

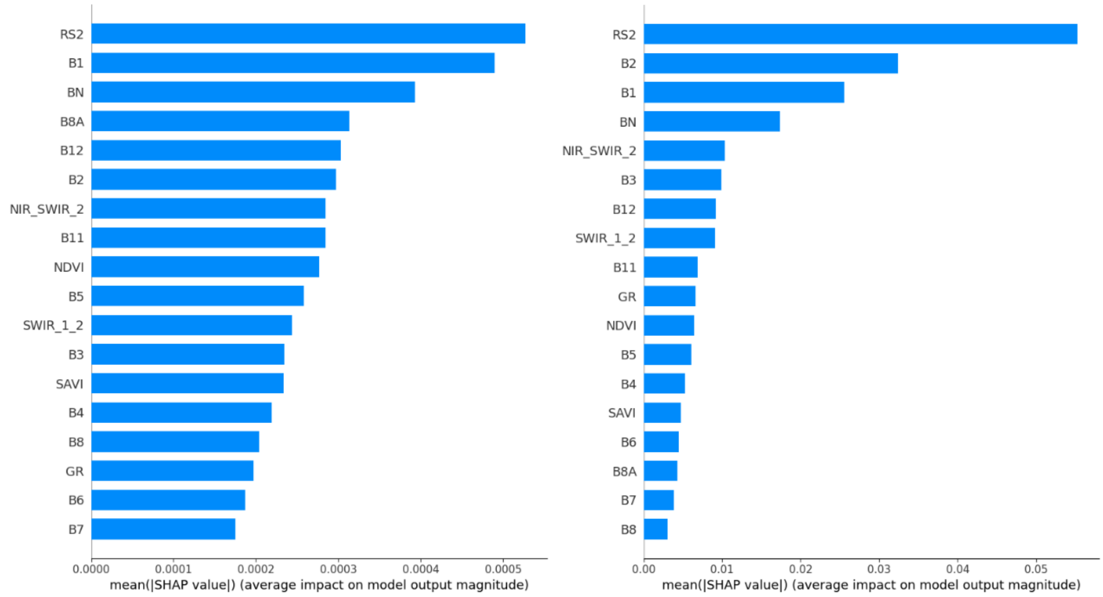


Figure S3. Mean SHAP feature importance's for cropland (left) and built-up (right) classes, sorted from highest to lowest. Both classes show RS2 as the highest predictor and BN follows with very high distinction.

Table S12. 2016-2021 Land change with area-reference estimators and error-adjusted standard errors expressed in 95% confidence intervals.

| | | Gross Cropland Change (ha) | | | Gross Built Up Change (ha) | | | Gross Forest Change (ha) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | *Gain* | *Loss* | *Net Change* | *Gain* | *Loss* | *Net Change* | *Gain* | *Loss* | *Net change* |
| 1. Rural | Srunen | 26.01 ± 1.19 | 21.54 ± 0.98 | 4.47 ± 5.46 | 37.33 ± 2.78 | 29.21 ± 2.18 | 8.12 ± 5.2 | 34.59 ± 3.27 | 47.19 ± 4.46 | -12.60 ± 4.93 |
| 2. Rural | Gading | 22.73 ± 1.04 | 21.18 ± 0.97 | 1.55 ± 5.48 | 24.46 ± 1.82 | 26.38 ± 1.97 | -1.92 ± 5.31 | 30.03 ± 2.84 | 29.66 ± 2.80 | 0.37 ± 5.13 |
| 3. Rural | Banjarsari | 21.08 ± 0.96 | 19.26 ± 0.88 | 1.83 ± 5.49 | 24.55 ± 1.83 | 18.53 ± 1.38 | 6.02 ± 5.37 | 27.38 ± 2.59 | 35.23 ± 3.33 | -7.85 ± 5.11 |
| 4. Urban | Gondang | 15.97 ± 0.73 | 15.79 ± 0.72 | 0.18 ± 5.53 | 22.18 ± 1.65 | 22.18 ± 1.65 | 0.00 ± 5.36 | 26.65 ± 2.52 | 26.84 ± 2.54 | -0.18 ± 5.19 |
| 5. Urban | Jetis Sumur | 29.94 ± 1.37 | 20.17 ± 0.92 | 9.77 ± 5.45 | 30.39 ± 2.27 | 26.29 ± 1.96 | 4.11 ± 5.27 | 35.14 ± 3.32 | 49.02 ± 4.63 | -13.87 ± 4.9 |
| 6. Urban | Pagerjurang | 15.52 ± 0.71 | 17.34 ± 0.79 | -1.83 ± 5.52 | 29.39 ± 2.19 | 27.20 ± 2.03 | 2.19 ± 5.27 | 32.86 ± 3.11 | 33.22 ± 3.14 | -0.37 ± 5.07 |
| 7. Urban | Karang K. | 16.61 ± 0.76 | 18.62 ± 0.85 | -2.01 ± 5.51 | 49.20 ± 3.67 | 36.60 ± 2.73 | 12.60 ± 5.06 | 42.26 ± 4.00 | 52.85 ± 5.00 | -10.59 ± 4.8 |
| 8. Rural | Not Affected | 28.57 ± 1.30 | 24.55 ± 1.12 | 4.02 ± 5.44 | 35.51 ± 2.65 | 31.13 ± 2.32 | 4.38 ± 5.2 | 41.62 ± 3.93 | 50.02 ± 4.73 | -8.40 ± 4.83 |
| 9. Urban | Not Affected | 29.48 ± 1.35 | 28.20 ± 1.29 | 1.28 ± 8.18 | 51.30 ± 3.82 | 41.99 ± 3.13 | 9.31 ± 7.56 | 55.77 ± 5.27 | 66.36 ± 6.27 | -10.59 ± 6.83 |
| Total Affected | | 147.87 ± 6.75 | 133.90 ± 6.11 | 13.97 ± 8.84 | 217.51 ± 16.22 | 186.39 ± 13.90 | 31.13 ± 11.09 | 228.92 ± 21.64 | 274.01 ± 25.90 | -45.09 ± 13.93 |

Note: 95% Confidence intervals for relative net change values consider the sum of squares for gain/loss estimates.

Sub-section 5.6.2
Not Affected (NA)
Affected (AF)

1. Cropland

Mean AF = 2.00 ha (43%)
Mean NA =2.65 ha (57%)
Total = 4.65 ha (100%)

0.57*(0.43) = 0.2451
Urban Dynamics = 24.5%

0.43 – 0.2451 = 0.185
Effects = 18.5%

2. Forest

Mean AF = -6.44 ha (40.4%)

Mean NA = -9.50 ha (59.6%)
Total = 15.94 ha (100%)

0.596 x 0.404 = 0.24
Urban Dynamics = 24%

0.404 – 0.24 = 0.164
Effects = 16.4%

3. Built-up

Mean AF = 4.45 ha (39.4%)
Mean NA = 6.85 ha (60.6%)
Total = 11.30 ha (100%)

0.606 x 0.394 = 0.239
Urban Dynamics = 23.9%

0.394 – 0.239 = 0.155
Effects = 15.5%